

GoalOS: The Validated Skill Graph

Reusable Capability, Chronicle-Gated Memory, and Proof-Carrying Skill Accumulation



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Montreal.AI Research | July 2026

Thesis. The durable asset in an autonomous proof economy is not a prompt, a transcript, or a polished model answer. It is a *validated skill*: a scoped, replayable, evidence-backed capability object that has survived proof, validation, risk checks, and CHRONICLE admission. The VALIDATED SKILL GRAPH is the institutional memory layer that stores and compounds those capabilities.

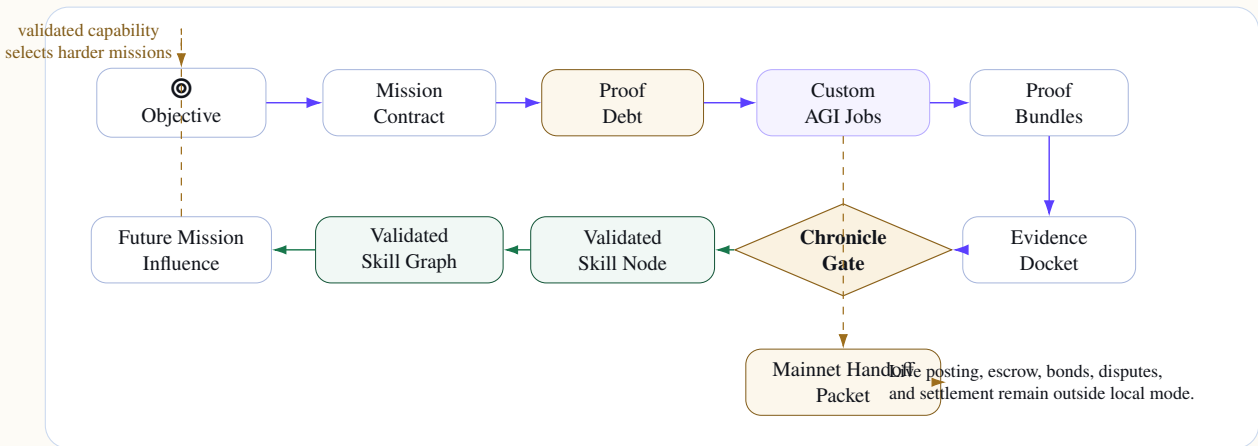


Figure 1: The VALIDATED SKILL GRAPH transforms validated outcomes into reusable capability. Drafting creates candidates; proof work creates evidence; CHRONICLE admits only validated skill.

1 ABSTRACT

GOALOS treats language-model output as candidate material, not institutional memory. A model can propose a plan, draft an argument, or produce a workflow, but those artifacts should not automatically shape future behavior. This paper isolates one layer of the GOALOS architecture: the VALIDATED SKILL GRAPH, a governed memory substrate for reusable capability. The graph stores only skills that have passed an explicit admission process: proof debt is retired, PROOFBUNDLES are returned, an EVIDENCE DOCKET aggregates and tests the evidence, the CHRONICLE Gate evaluates the configured proof level, and only then does a capability become reusable memory. The resulting graph is not a prompt library. It is a typed, versioned, scoped, replayable, provenance-backed network of capability objects. We formalize the skill-node object model, edge semantics, proof-level-dependent admission, repair paths through custom AGI JOBS, skill composition, worth-keeping utility, threat model, evaluation program, and commercial implications. The central contribution is a memory discipline for autonomous work: what compounds is not output volume, but validated reusable capability.

2 INTRODUCTION

The first wave of AI products made generation cheap. They produced answers, drafts, code, plans, reports, and summaries. But institutions cannot govern themselves with persuasive text alone. They require a reliable answer to a harder question: which outputs deserve reuse, trust, propagation, settlement, and future influence?

The VALIDATED SKILL GRAPH answers that question by separating *candidate output* from *validated capability*. A skill is admitted only after an evidence path exists. A reusable capability is not a sentence, a prompt, or a transcript; it is a proof-carrying operational object with provenance, scope, acceptance tests, replay instructions, risk boundaries, and measured utility.

This paper focuses only on that layer. It assumes the broader GOALOS operating loop: objective, mission contract, autonomous work, verification, EVIDENCE DOCKET, governed decision state, action graph, CHRONICLE, and reusable capability. Mission OS defines this as a proof-to-action system in which the user supplies an objective and GOALOS runs until proof is done [1]. AEP-001 states the constitutional boundary: private intelligence remains off-chain, while proof commitments, attestations, and evolution rights can be governed; no proof, no evolution; no evaluation, no propagation; no rollback, no release [2].

Positioning. A prompt repository stores instructions. A transcript archive stores conversation. A vector database stores recallable fragments. The VALIDATED SKILL GRAPH stores validated capability: what worked, under which constraints, with which evidence, for which scope, and with which proof level.

3 PROBLEM: MEMORY WITHOUT PROOF

Autonomous systems become dangerous or unreliable when memory is too permissive. A system that remembers every confident output can compound errors, overclaims, stale evidence, reward hacks, and unsupported assumptions. This creates four failure modes.

- F1. Contagious unsupported claims.** A persuasive answer enters memory and influences later missions without proof.
- F2. Prompt-library drift.** Useful instructions are copied and reused outside the context where they were valid.
- F3. Invisible provenance loss.** Future operators cannot determine what evidence justified a prior capability.
- F4. Ungated self-improvement.** The system treats successful-sounding behavior as improvement, even when it has not passed replay, risk, or validation checks.

The VALIDATED SKILL GRAPH is a countermeasure. It creates a narrow bridge from output to memory. Nothing becomes durable merely because it is well-written. Memory is earned through evidence.

4 DEFINITIONS

Definition 1 (Candidate Work). *Candidate work is any draft, plan, code fragment, analysis, workflow, claim, artifact, or agent trace produced before CHRONICLE admission. Candidate work may be useful, but it has no durable memory authority.*

Definition 2 (Proof Debt). *Proof debt is an explicit record of missing evidence required before a claim, artifact, or capability can be trusted at the selected proof level. Proof debt is retired by evidence, review, replay, test results, or other accepted proof artifacts.*

Definition 3 (Validated Skill). *A validated skill is a scoped capability object admitted by the CHRONICLE Gate after its proof debt is sufficiently retired. A validated skill includes scope, preconditions, evidence, replay instructions, risk boundaries,*

version metadata, and usage policy.

Definition 4 (Validated Skill Graph). The *VALIDATED SKILL GRAPH* is a typed directed multigraph whose nodes are validated skills and whose edges encode reuse, composition, teaching, packaging, audit, repair, retirement, and future mission influence.

5 SYSTEM POSITIONING

The *VALIDATED SKILL GRAPH* lives at the boundary between proof and memory. Figure 2 shows the governed transition from candidate work to reusable capability.

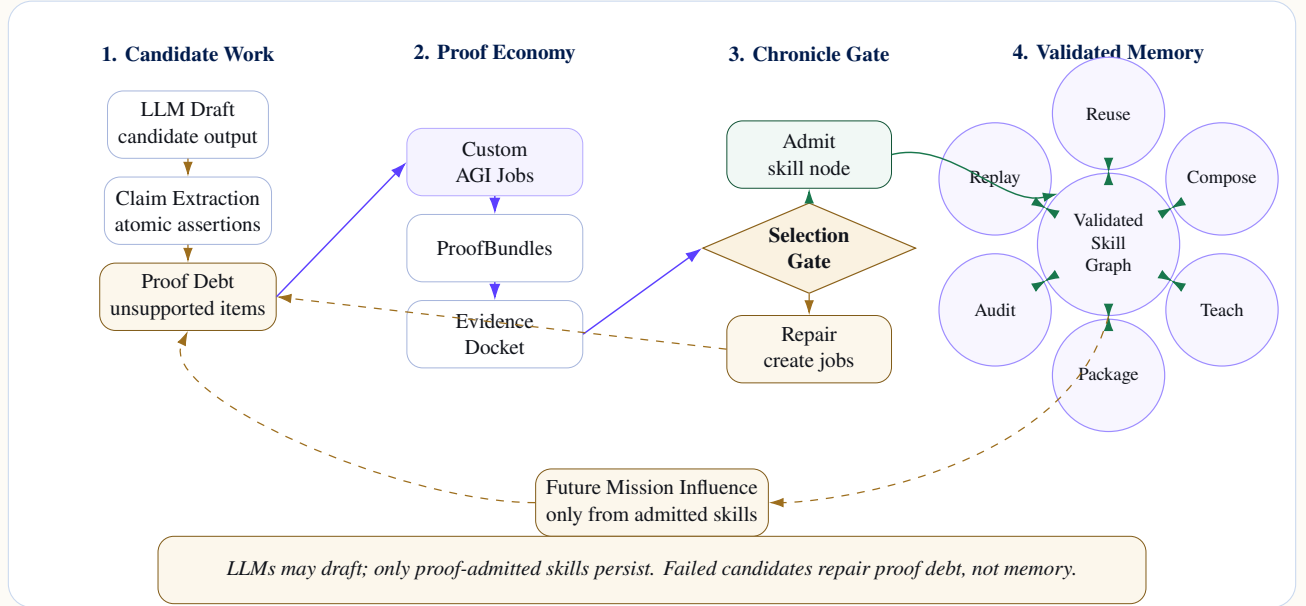


Figure 2: *VALIDATED SKILL GRAPH* as a governed memory substrate. Candidate output becomes durable capability only after proof and *CHRONICLE* admission.

6 FORMAL OBJECT MODEL

Let a skill node be represented by

$$s = (id, type, scope, L, P, E, V, R, B, U, \tau, \rho), \quad (1)$$

where id is a stable identifier, $type$ is the skill class, $scope$ is its permitted operating domain, L is the proof level, P is the proof-debt ledger, E is the evidence set, V is validator coverage, R is replay material, B is the risk boundary, U is utility telemetry, τ is version and lineage metadata, and ρ is the rollback or retirement policy.

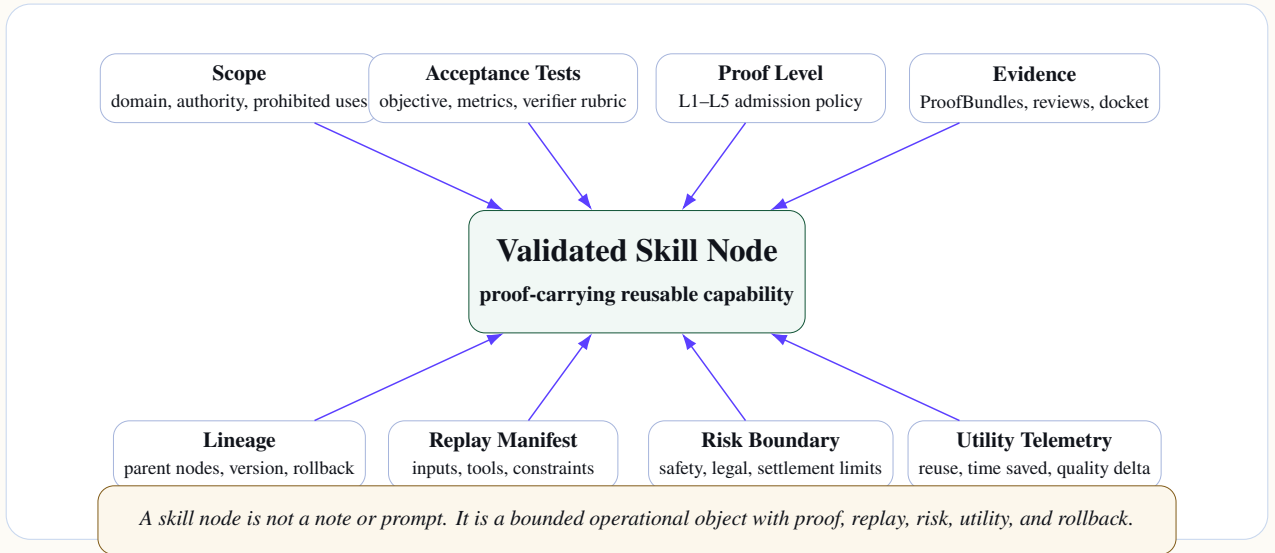


Figure 3: Skill-node passport. Every admitted node carries operational identity, evidence, replay, risk, utility, and rollback metadata.

Table 1: Canonical skill-node fields.

Field	Function
Scope	Defines where the capability may be reused; prevents domain drift.
Proof level	Binds the node to an admission standard from exploratory through formal.
Evidence	Links to PROOFBUNDLES, EVIDENCE DOCKET entries, reviews, replay logs, and source commitments.
Acceptance tests	Specifies what must be true before the node is considered valid.
Validator coverage	Records reviewer or AGI Node attestations and unresolved dissent.
Replay manifest	Enables reproduction or reconstruction of the capability path.
Risk boundary	States safety, legal, policy, settlement, and operational constraints.
Utility telemetry	Records observed reuse, quality, time saved, error rate, and mission impact.
Rollback policy	Specifies retirement, repair, quarantine, and version reversion triggers.

7 CHRONICLE ADMISSION POLICY

The CHRONICLE Gate is the memory firewall. It decides whether a candidate skill becomes durable capability, requires repair, or is rejected. Let $A_L(s) \in \{0, 1\}$ denote admission at proof level L . A simple admission rule is

$$A_L(s) = 1 \iff \begin{cases} \text{support}(s) \geq \theta_L^{\text{support}}, \\ \text{replay}(s) \geq \theta_L^{\text{replay}}, \\ \text{validation}(s) \geq \theta_L^{\text{validation}}, \\ \text{risk}(s) \leq \theta_L^{\text{risk}}, \\ \text{scope_ok}(s) = 1. \end{cases} \quad (2)$$

The thresholds are policy parameters. Raising proof level increases the evidence burden and reduces the chance that persuasive but ungrounded material enters memory.

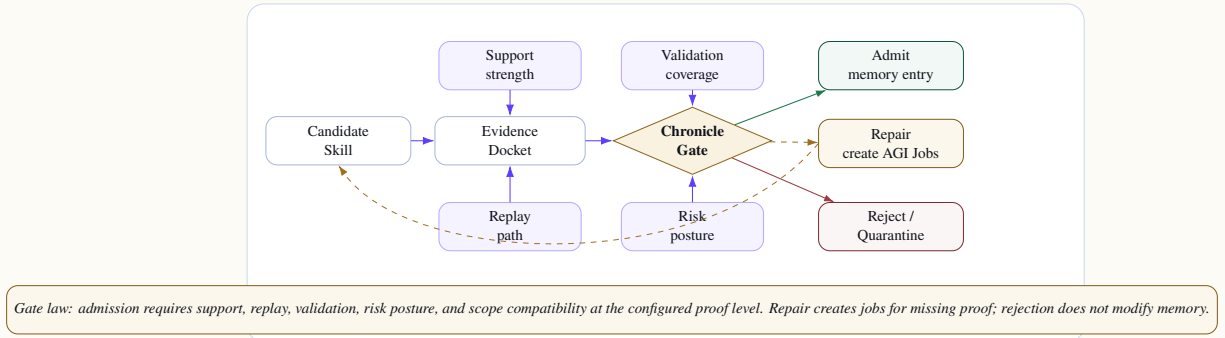


Figure 4: Chronicle admission policy. Evidence becomes authority only when it satisfies the selected proof level.

8 FLEXIBLE PROOF LEVELS

Different work requires different proof. A brainstorming session, an internal playbook, a public claim, a mainnet handoff, and a production-grade memory entry should not share the same admission threshold. GOALOS therefore treats proof level as a first-class mission parameter.

Table 2: Proof levels for skill admission.

Level	Name	Required evidence posture	Memory effect
L1	Structural / Demo	Mission structure, explicit proof debt, local artifact coherence.	No durable CHRONICLE influence.
L2	Internal Review	Operator confirms usefulness, scope, and risk boundaries.	Temporary reuse, clearly labeled.
L3	External Review	Independent reviewer, AGI Node validation, or cross-source corroboration.	Scoped reusable skill.
L4	Mainnet Handoff	Validated evidence, job specification, settlement boundary, and explicit handoff packet.	Skill may inform external execution preparation.
L5	Chronicle / Formal	Multi-validator evidence, replay, baseline comparison, risk gate, and rollback condition.	Durable skill with future mission influence.

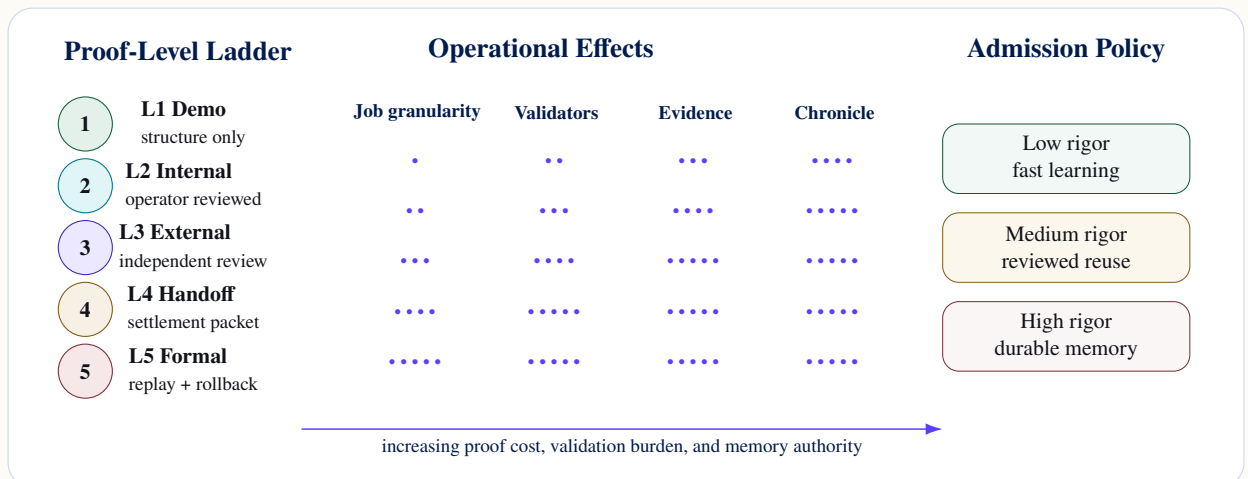


Figure 5: Proof-level policy. The selected level calibrates job design, validator burden, evidence strictness, and memory authority.

9 GRAPH SEMANTICS

The VALIDATED SKILL GRAPH is a directed typed multigraph $G_t = (S_t, E_t)$. Each edge is a tuple

$$e = (s_i, \rho, s_j, w, C, \pi), \quad (3)$$

where s_i and s_j are skill nodes, ρ is the relation type, w is an observed utility or confidence weight, C is the context in which the relation has been observed, and π is evidence provenance.

Table 3: Core graph relations.

Relation	Meaning	Example
Reuse	Skill applies in a new mission context.	A reviewer-room pattern used for a new AI vendor diligence mission.
Compose	Skills combine into a higher-order capability.	Claim-boundary skill + job-design skill = proof-mission factory.
Teach	Skill can train operators or agents.	A validated review heuristic becomes a Masterclass lesson.
Package	Skill is converted into a product artifact.	A proof packet becomes an AI Trust Room template.
Audit	Skill requires periodic review or risk checks.	A compliance-related skill is revalidated when policies change.
Retire	Skill is deprecated, superseded, or quarantined.	A skill fails replay or becomes unsafe outside its original scope.
Influence	Skill reduces uncertainty in future mission planning.	Validated evidence reduces proof debt in a related mission.

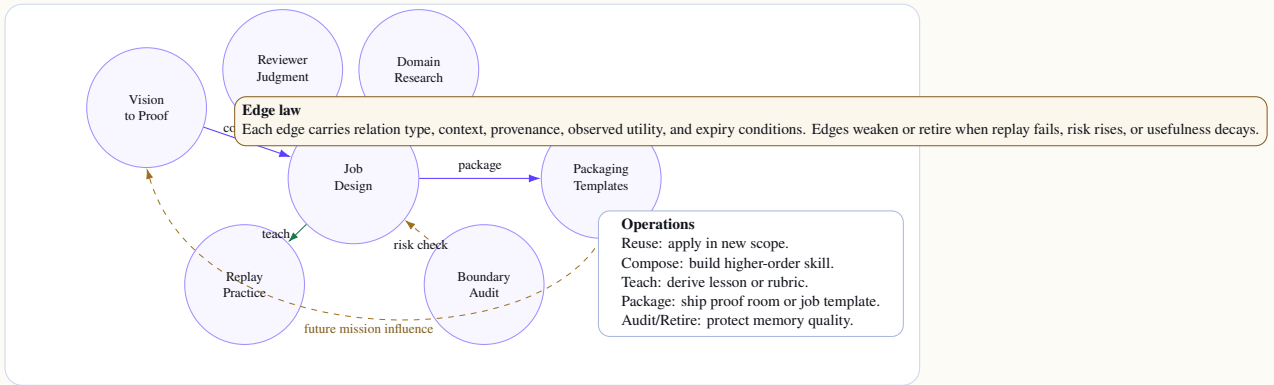


Figure 6: Graph semantics. Validated skills are connected by typed, evidence-carrying operations rather than informal similarity.

10 SKILL CAPTURE STUDIO

The Skill Capture Studio converts expert judgment into candidate skills. The objective is not to memorize a meeting or imitate an expert style; it is to isolate reusable operations that can be validated.

- S1. Extract episodes.** Identify concrete expert moves: framing, critique, boundary-setting, repair, evaluation, packaging.
- S2. Create candidate nodes.** Assign type, scope, proof level, and first-pass utility hypothesis.
- S3. Generate proof debt.** Record what evidence would be needed before reuse is safe.
- S4. Create custom jobs.** Convert unresolved proof into AGI JOBS with worker roles, validators, and acceptance tests.
- S5. Admit or repair.** Use the CHRONICLE Gate to determine whether the capability becomes durable memory.

11 WORTH-KEEPING UTILITY

Not every validated artifact should become a durable skill. Memory can bloat. The graph therefore needs a worth-keeping function that trades off value, reuse, cost, risk, and maintenance burden. One practical score is

$$W(s) = \alpha V(s) + \beta R_u(s) + \gamma Q(s) - \lambda C_p(s) - \mu Risk(s) - \nu M(s), \quad (4)$$

where V is observed value, R_u is reuse potential, Q is quality or validation strength, C_p is proof cost, $Risk$ is residual risk, and M is maintenance burden. A skill is retained only if

$$A_L(s) = 1 \quad \text{and} \quad W(s) > \theta_{type,scope}^{keep}. \quad (5)$$

Worth-keeping principle. A skill should enter durable memory only when its expected future utility exceeds its proof cost, risk, and maintenance burden. Passing proof is necessary; being worth keeping is an additional economic and governance test.

12 CAPABILITY COMPOSITION

Skill composition is powerful because it turns atomic capabilities into higher-order capabilities. It is also dangerous because composition may amplify risk. Therefore two or more skills may compose only when their scopes, assumptions, evidence levels, and risk boundaries are compatible.

Proposition 1 (Composition Gate). Let $S_C = \{s_1, \dots, s_n\}$ be a candidate composition. The composite skill s_C may enter the graph only if each component is admitted, the composition proof level is at least the maximum required level of the output scope, and the joint risk remains under policy threshold:

$$A(s_C) = 1 \Rightarrow \bigwedge_{i=1}^n A(s_i) = 1, \quad L(s_C) \geq \max_i L(s_i), \quad \text{Risk}(S_C) \leq \theta_C. \quad (6)$$

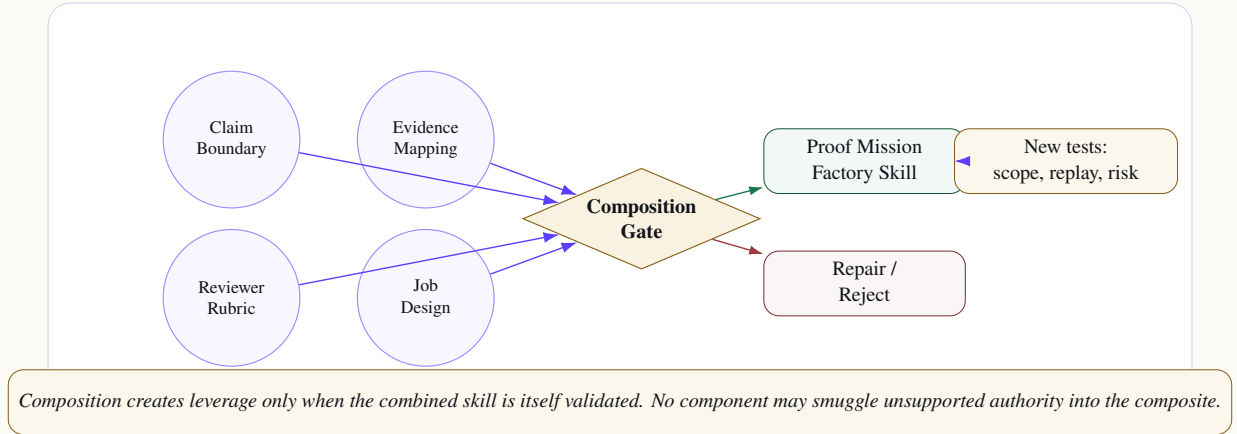


Figure 7: Capability composition gate. Higher-order skills inherit evidence obligations and risk boundaries from their components.

13 CUSTOM AGI JOBS FOR SKILL REPAIR AND EXTENSION

A failed skill candidate should not vanish silently. The repair path converts missing evidence into custom AGI JOBS. The job factory starts empty. Jobs are generated only after a mission objective, candidate skill, proof level, and proof-debt ledger are known.

Input: Candidate skill s , proof level L , proof debt ledger $P(s)$

Output: Custom AGI Job queue J_s

$J_s \leftarrow \emptyset$

foreach debt item $p \in P(s)$ **do**

determine required evidence class at level L
 select worker role and validator role
 compile acceptance tests and blocked claims
 construct ProofBundle return path
 append job specification to J_s

end

return J_s

Algorithm 1: Skill repair job generation.

14 MAINNET HANDOFF BOUNDARY

The VALIDATED SKILL GRAPH can inform settlement preparation, but local memory admission is not on-chain settlement. A local system may prepare a handoff packet containing mission scope, evidence commitments, proof status, and job specifications. Live posting, signing, escrow, bonds, validator selection, disputes, and settlement occur only in the external live system. This boundary is essential because an AGIJobManager-like protocol involves job posting, assignment, escrow,

bonds, validation, disputes, and settlement risk [4].

Boundary. Local GOALOS can prepare proof and handoff packets. It does not move funds, create on-chain records, settle disputes, slash validators, or grant production authority. Crossing the boundary requires explicit operator action and the live protocol’s rules.

15 EVALUATION PROGRAM

A VALIDATED SKILL GRAPH should be judged by whether it improves future work without increasing ungrounded authority. We propose the following metrics.

Table 4: *Evaluation metrics for validated-skill accumulation.*

Metric	Measurement
Admission precision	Fraction of admitted skills that later pass replay, review, and reuse tests.
Admission recall	Fraction of useful candidate skills that the gate admits after appropriate repair.
Proof cost per durable skill	Total proof, validation, and review cost divided by accepted durable skill count.
Reuse lift	Improvement in time, cost, or quality for missions using prior validated skills versus a no-graph baseline.
Composition success	Fraction of composite skills that deliver value without increased risk or drift.
Memory bloat rate	Growth of low-utility or never-reused skills; should remain bounded.
Rollback correctness	Speed and completeness of skill retirement after replay failure or policy change.
Boundary integrity	Absence of local-to-live settlement confusion, unauthorized elevation, or unsupported mainnet claims.

16 THREAT MODEL

The VALIDATED SKILL GRAPH is exposed to adversarial and operational risks.

Table 5: *Threats and mitigations.*

Threat	Failure mode	Mitigation
Evidence laundering	Weak evidence is repackaged as proof.	Evidence provenance, validator coverage, replay requirements, and negative attestations.
Prompt contagion	Unverified prompt patterns influence future missions.	Chronicle admission only after proof; candidate work quarantined.
Scope drift	A valid skill is reused outside its original context.	Scope signatures, compatibility checks, and warning labels.
Validator capture	Reviewers collude or approve weak proof.	Multi-reviewer coverage, reputation, dissent recording, and challenge windows.
Graph bloat	Too many low-value skills accumulate.	Worth-keeping score, decay, retirement, and usage telemetry.
Composition risk	Combined skills create new unsupported authority.	Composition Gate, composite proof level, and joint risk evaluation.
Replay failure	A skill cannot be reproduced under stated conditions.	Replay manifests, artifact hashes, dependency records, and rollback policy.
Settlement confusion	Local proof is mistaken for on-chain settlement.	Explicit local/live boundary, handoff-only status, and operator confirmation.

17 FALSIFICATION TESTS

The VALIDATED SKILL GRAPH claim is falsifiable. The system fails if any of the following persist after repair.

- T1. Skills admitted at a given proof level fail replay at unacceptable rates.
- T2. Future missions using the graph do not outperform comparable missions without the graph.
- T3. The graph admits unsupported claims because they sound authoritative.
- T4. Skill reuse increases risk, policy violations, or settlement-boundary confusion.
- T5. The system cannot explain why a skill was admitted, repaired, rejected, retired, or reused.
- T6. Memory grows faster than validated utility, creating bloat rather than leverage.

18 COMMERCIAL IMPLICATIONS

The VALIDATED SKILL GRAPH reframes the business model. The product is not access to prompts. The product is a mechanism for converting expert judgment and proof-bearing work into reusable capability. In a practical deployment, the business layers are:

- B1. **Proof Mission Factory.** Convert high-value objectives into bounded missions, proof debt, and validated artifacts.
- B2. **Skill Capture Studio.** Convert expert judgment into candidate skills and proof jobs.
- B3. **AI Trust Room Product Line.** Package validated capabilities into buyer-facing proof rooms.
- B4. **Enterprise Proof OS.** Manage policy, evidence, validators, and CHRONICLE admission at organizational scale.
- B5. **AGI Jobs Validator Marketplace.** Route unresolved proof work to agents and reviewers under explicit validation rules.

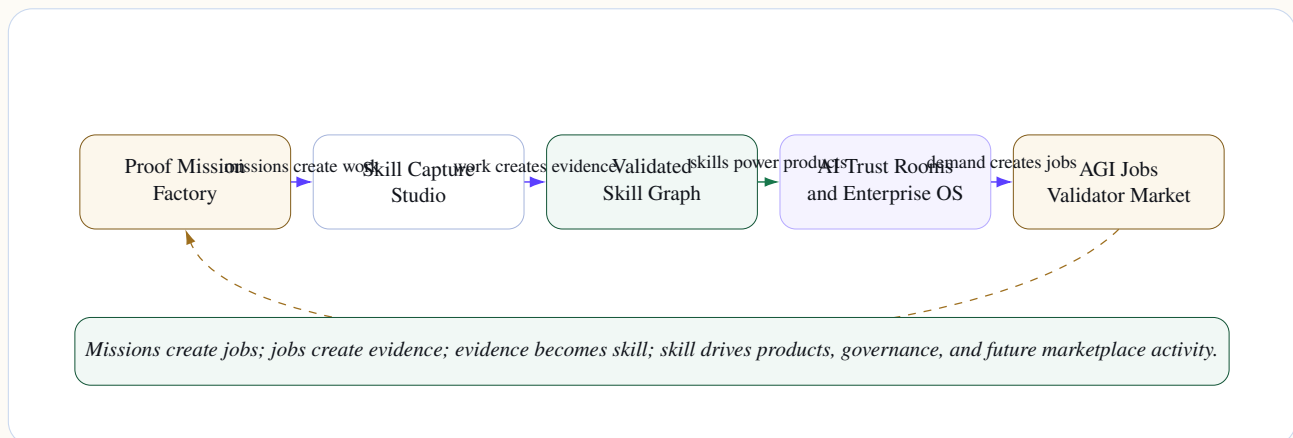


Figure 8: Validated skill commercial flywheel. The durable compounding asset is the graph of proven capability.

19 CONCLUSION

The VALIDATED SKILL GRAPH is the memory discipline of a proof economy. It prevents language-model output from becoming durable authority merely because it is articulate, plausible, or convenient. It admits only scoped, evidence-backed, replay-aware, risk-bounded capability. It supports reuse, composition, teaching, packaging, audit, and retirement. It makes skill accumulation measurable and governable.

The core result is simple: the durable asset is not the prompt. It is the proof-carrying skill node. The durable advantage is not more output. It is a graph of validated capability that can be reused, composed, taught, packaged, audited, retired, and trusted.

A CANONICAL SKILL NODE JSON

```

{
  "skill_id": "vsg.skill.vision_to_proof.001",
  "type": "vision_to_proof",
  "scope": {
    "domain": "AI Trust Room / proof mission design",
    "allowed_use": ["mission framing", "claim boundary", "proof debt extraction"],
    "blocked_use": ["legal certification", "security certification", "investment claim"]
  },
  "proof_level": "L3_external_review",

```

```

"evidence": {
  "proofbundles": ["pb://..."],
  "evidence_docket": "ed://...",
  "reviewer_attestations": ["attest://..."],
  "replay_manifest": "replay://..."
},
"acceptance_tests": [
  "claims explicit",
  "proof debt enumerated",
  "review route attached",
  "blocked claims visible"
],
"utility": {
  "reuse_count": 7,
  "median_time_saved_minutes": 42,
  "quality_delta": 0.18,
  "risk_incidents": 0
},
"chronicle_status": "admitted",
"version": "1.2.0",
"rollback": "retire if replay fails twice or scope policy changes"
}

```

B MINIMAL ADMISSION ALGORITHM

Input: Candidate skill s , proof level L , Evidence Docket D

Output: admit, repair, reject, or quarantine

compute support, replay, validation, risk, and scope scores from D

if *scope violation or prohibited claim* **then**

 | **return** *quarantine*

end

if *scores meet thresholds for L and worth-keeping utility is positive* **then**

 | **return** *admit*

end

if *missing evidence can be converted into jobs* **then**

 | **return** *repair*

end

return *reject*

Algorithm 2: Chronicle admission decision.

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